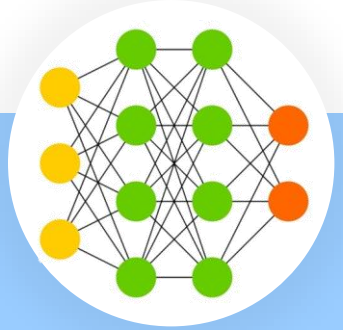
香港教育大學
The Education University of Hong Kong

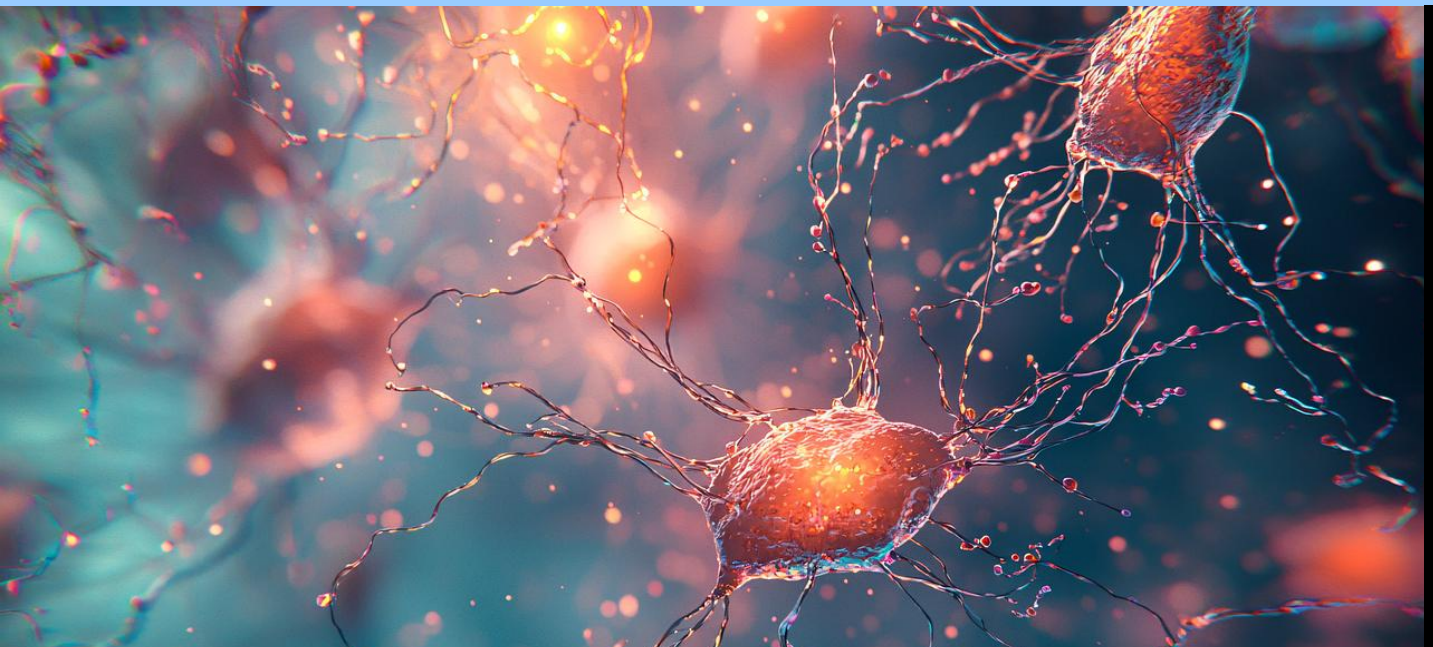
MSc(AIET) 2025-26

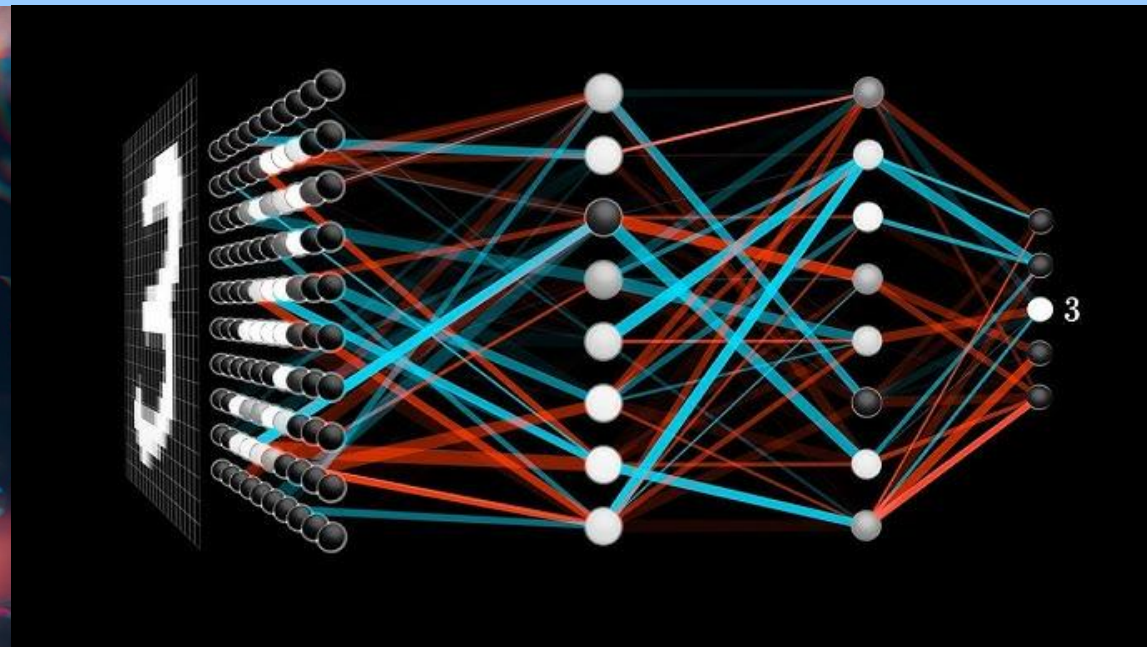


MTH6184 Data Mining in STEM Education

Chapter 9. Neural Networks

Instructor: Philip Yu

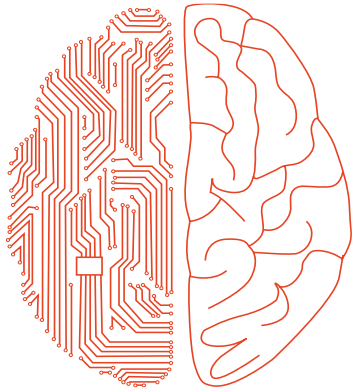




Chapter Outcomes

After completing this chapter, you are able to:

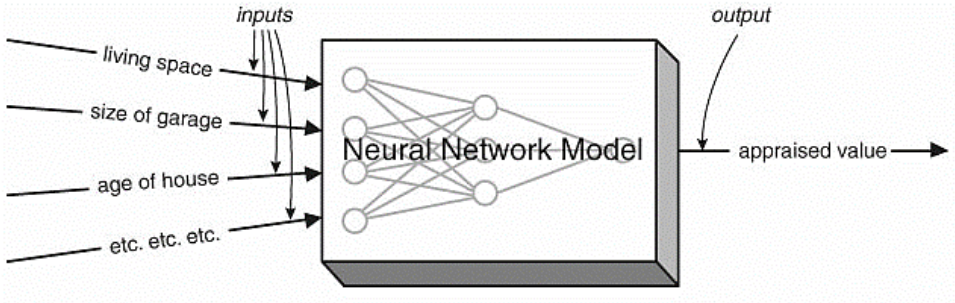
- Compare human neural network and artificial neural network
- Understanding Neural Networks with TensorFlow Playground
- Understand the relation between a generalized linear model and a neural network
- Forecast a time series using neural networks
- How to avoid overfitting in neural networks



Examples

- Stock Price Forecasting using deep neural networks
  - used 850 time series features (eg. technical indicators) as input for neural network to forecast 21-day stock return
  - https://neuravest.net/deep-neural-networks-to-forecast-stock-prices/
- Soccer Games Predictions
  - used neural network to predict the match outcome (1=Home-win, X=Tie, 0=Away-win)
  - https://www.prosocr.com/

- Property Valuation
  - combined forecasts based on hedonic method, repeat-sales method and neural network
  - https://www.homeinfomax.com/Information/Real_Estate_Records/valuepoint_information.php



Journal of Property Research, December 2008, **25**(4) 321–342
Routledge Taylor & Francis Group

An Artificial Neural Network and Entropy Model for Residential Property Price Forecasting in Hong Kong

K. C. LAM, C. Y. YU and K. Y. LAM
Department of Building and Construction, City University of Hong Kong, Hong Kong

Table 3. Dependent and independent variables

Variable types	Abbreviation	Variable names
Dependent variables	PRICE	Price
Independent variables (micro-economic factors)	FL	Number of floors
	AREA	Area (square feet)
	AGE	Age of the building
	DIST	District
	BA	Balcony
	DU	Duplex
	PF	Platform
	FR	Roof/flat roof
	PS	Primary sale
	GN	If or not owns garden(s)
	RS	Distance to railway station
	BED	Number of bedrooms
	BATH	Number of bathrooms
	LIFT	Number of lifts
	PSN	Number of primary school nearby
	SSN	Number of secondary school nearby
	CH	Clubhouse
	SP	If or not owns swimming pool(s)
	PG	Playground, basketball court etc.
	UBL	Units per block
	UFL	Units per floor
	UT	Total units
Independent variables (macro-economic factors)	GDP	Gross domestic product
	GNP	Gross national product
	BOP	Balance of payment
	EFIF	External factor income flows
	CCPI	CCPI
	UDM	Number of underemployment
	UEM	Number of unemployment

Artificial Neural Network Example

Health Care Management Science 1 (1998) 29–37

甲狀腺

An investigation of neural networks in thyroid function diagnosis

Guoqiang (Peter) Zhang [a] and Victor L. Berardi [b]

[a] *Department of Decision Sciences, College of Business Administration, Georgia State University, Atlanta, GA 30303, USA*
E-mail: gzhang@kentvm.kent.edu
[b] *Department of Management, College of Business, Bloomsburg University, Bloomsburg, PA 17815, USA*

- 7200 patients in 1985 and 1986
  - 3772 patients in 1985 (training)
  - 3428 patients in 1986 (testing)
- To determine whether a patient has
  - a normally functioning thyroid (Euthyroid) (92.6%),
  - an under-functioning thyroid (hypothyroid) (5.1%), or
  - an over-active thyroid (hyperthyroid) (2.3%)
- 21 predictors:
  - 15 binary variables
  - 6 continuous variables
  - Include age, sex, health condition, and results of medical tests

- Performance comparison between neural network with 10 hidden nodes and logistic regression

Class	Measure	Neural network		Logistic regression	
		Training	Test	Training	Test
Hyperthyroid	% Correct	95.70%	82.19%	0.00%	0.00%
	Correct #	89	60	0	0
Hypothyroid	% Correct	96.86%	94.92%	0.00%	0.00%
	Correct #	185	168	0	0
Euthyroid	% Correct	99.66%	98.71%	100.00%	100.00%
	Correct #	3476	3137	3488	3178
Overall	% Correct	99.42%	98.16%	92.47%	92.71%
	Correct #	3750	3365	3488	3178

- Neural network demonstrate superior accuracy in predicting abnormal thyroid functioning compared to logistic regression model.

Deep Neural Network in Language Learning

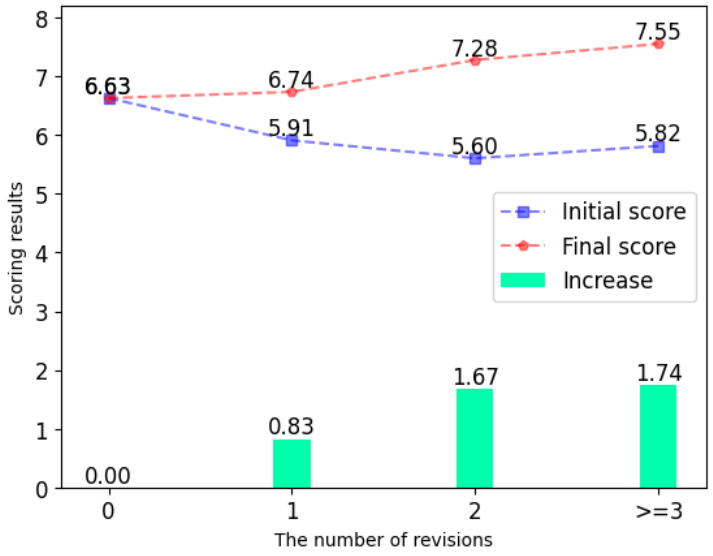
Students are asked to write a sentence to describe the picture.

Student A: “The boy is learning how to ride a bike.”

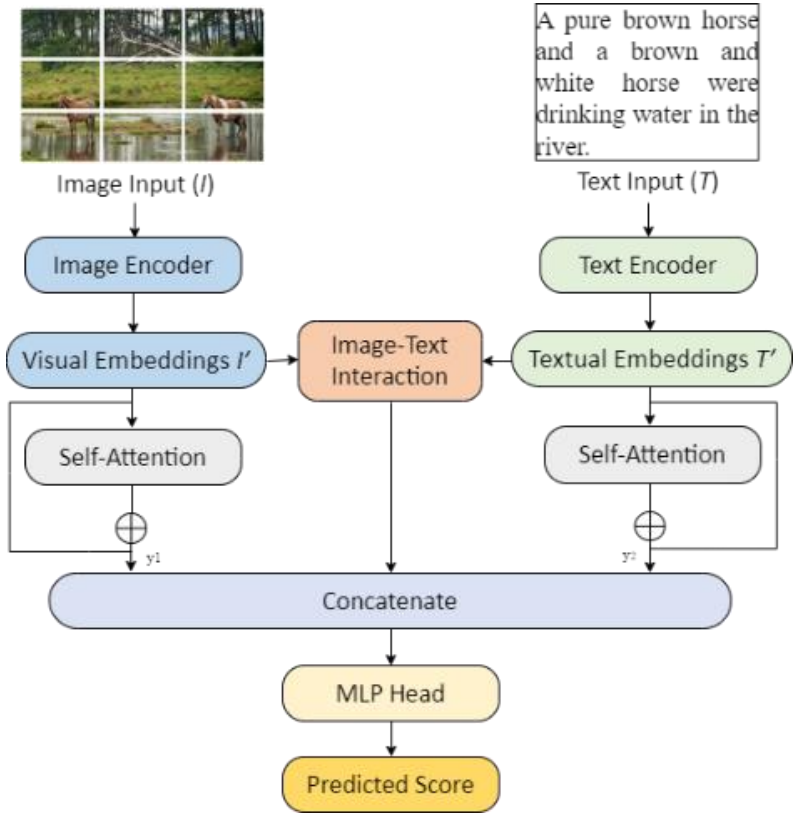
Student B: “On the grass, boy is trying to keep his balance on the bike, and his father is following him to ensure his safe.”

Which sentence should get a higher score in a scale of 0-10?

GenAI Feedback Help Students Progressively Improve Their Writing

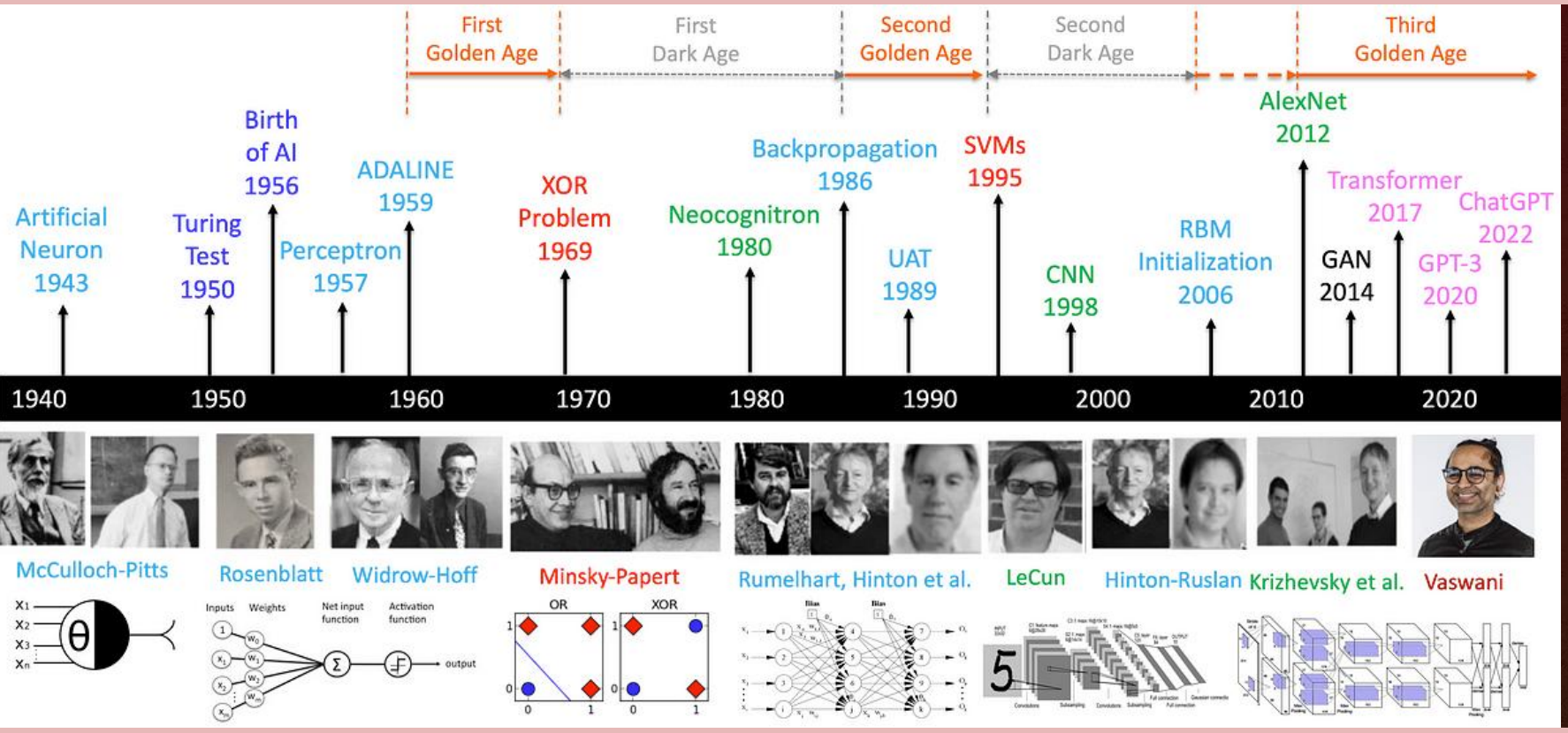


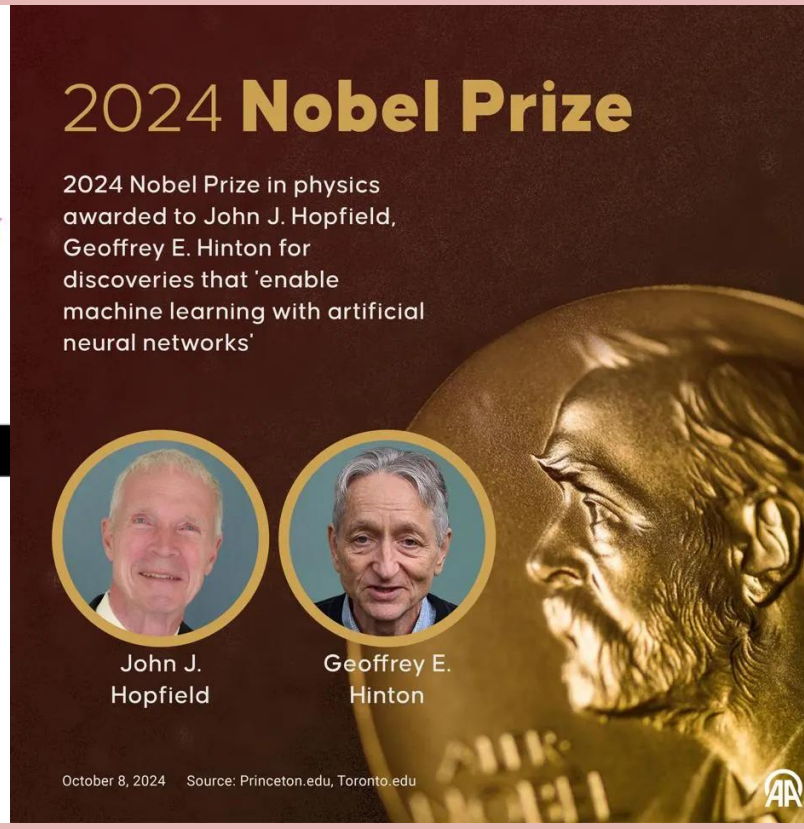
Our cross-modal deep learning achieved testing MAE=0.397 for scoring



Zhao, R., Zhuang, Y., Xie Z. and Yu, P.L.H. (2024). Facilitating self-directed language learning in real-life scene description tasks with automated evaluation. *Computers & Education*, 2024: 105106.

History of Artificial Neural Networks (ANN) and Deep Learning





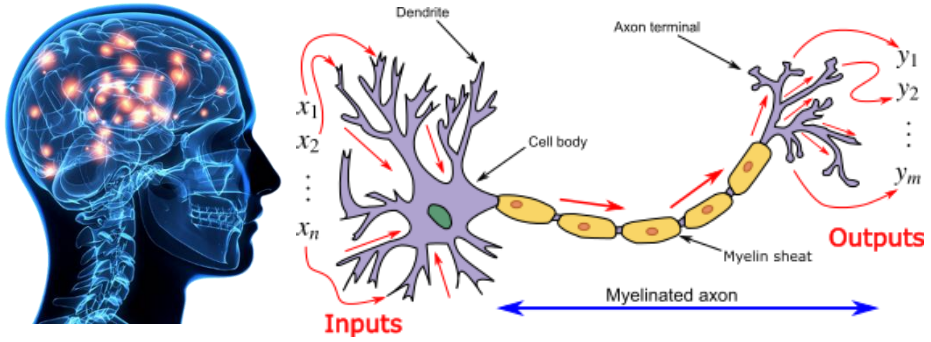
https://medium.com/@lmpo/a-brief-history-of-ai-with-deep-learning-26f7948bc87b

https://www.forbes.com/sites/geruiwang/2024/10/14/nobel-prizes-show-how-ai-is-shaping-history-5-reasons-you-should-care/

Introduction to Neural Network

- Human brain is the best learning unit in the universe as far as we know. So, for decades scientists and engineers try artificially recreate the process that human brain used to learn things quickly.
- Human brain consists of billions of interconnected neurons, transmitting signals between neurons.

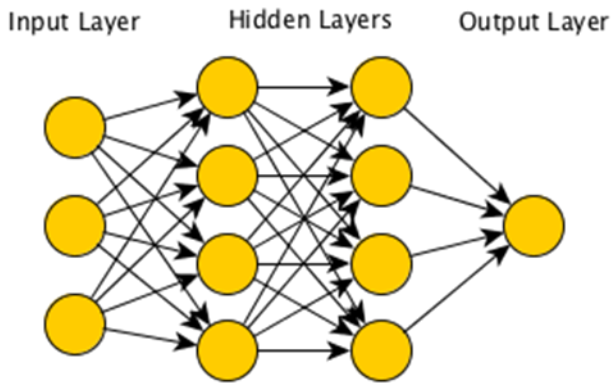
Human brain and neural network



https://en.wikipedia.org/wiki/Biological_neuron_model

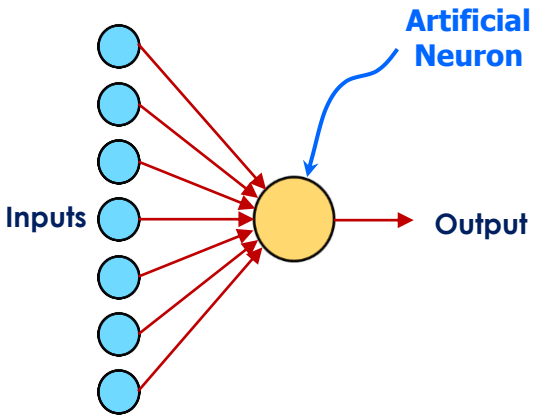
- Artificial neural network (ANN)
  - ANN is generally presented as systems of interconnected “neurons”.
  - The connections have numeric weights than can be tuned, making neural nets adaptive to inputs and capable of learning.

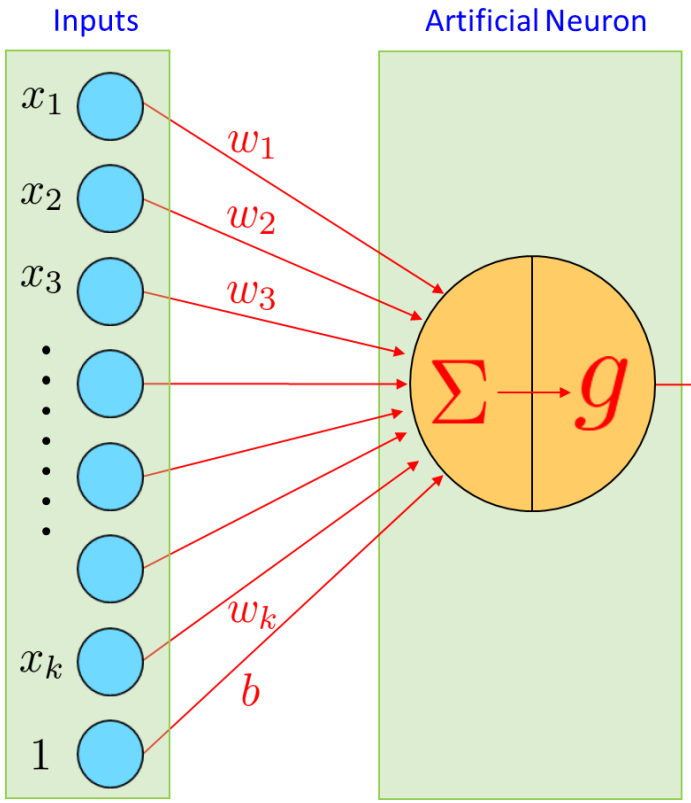
Artificial neural network



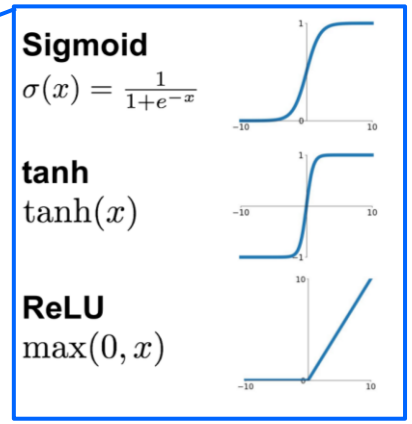
ANN: Neuron Model

- Each (artificial) neuron with k input features is known below:
  - Input features are weighted by model weights in $\boldsymbol{\omega}$ and bias (intercept) b.
  - The sum of the weighted inputs and the bias forms the input to the activation function g.
  - Neurons can use any differentiable activation function $g(\cdot)$ to generate the output.

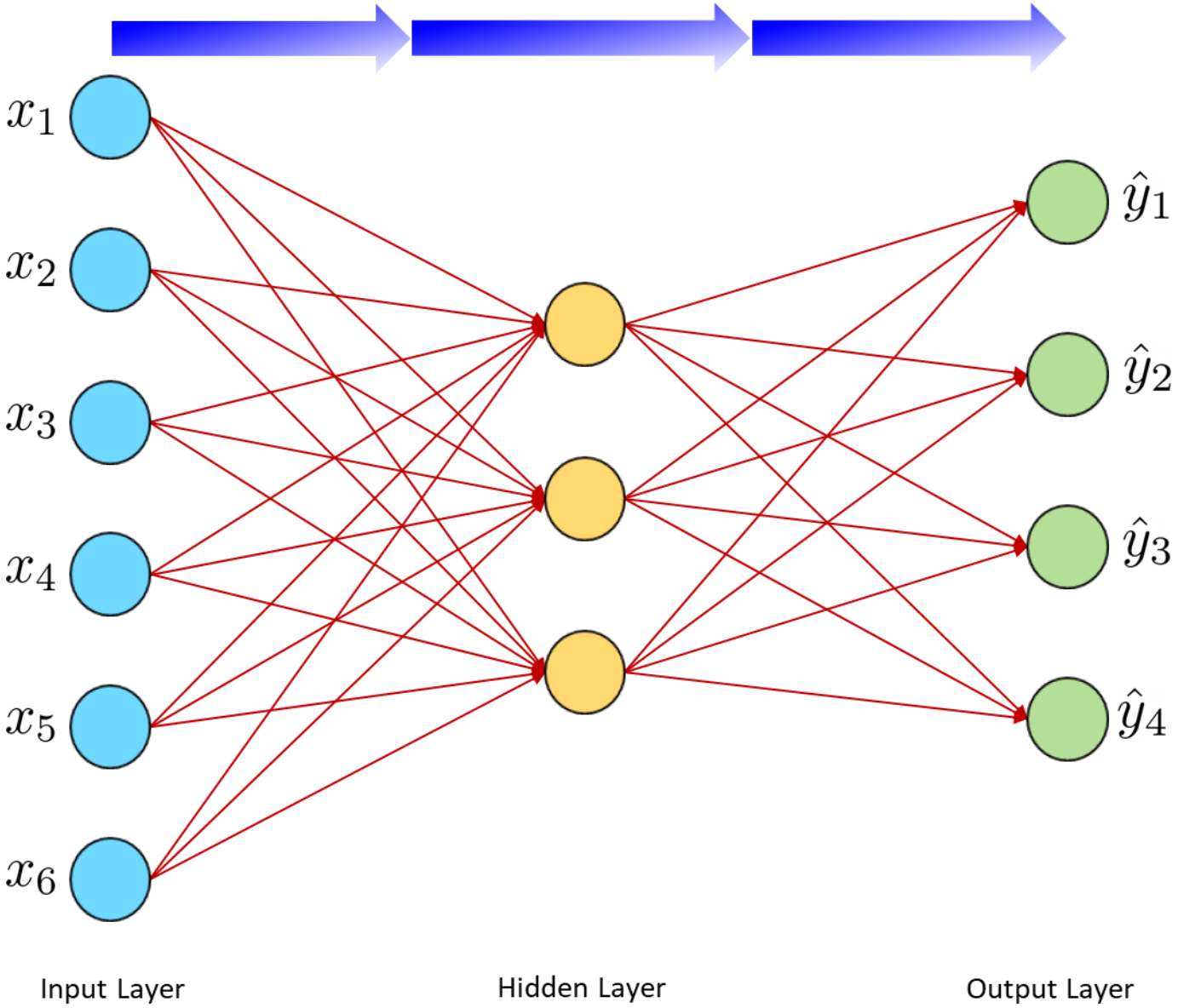




$$a = g(\boldsymbol{w}^T \boldsymbol{x} + b)$$
$$\boldsymbol{w}^T = (w_1, w_2, \ldots, w_k)$$
$$\boldsymbol{x}^T = (x_1, x_2, \ldots, x_k)$$



ANN: Feed-forward Model



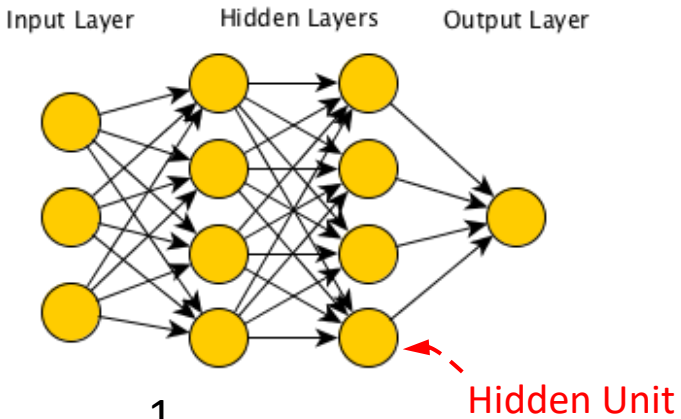
- In feed-forward neural network, the information moves in only one direction—forward—from the input nodes, through the hidden nodes (if any), and to the output nodes.
- There are no cycles or loops in the network.

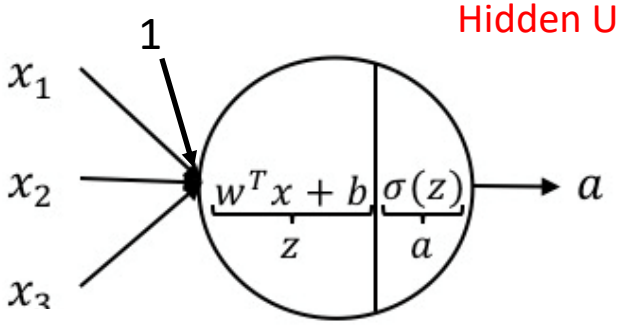
3 Key Elements in ANN

```
sklearn.neural_network.MLPClassifier / MLPRegressor:
hidden_layer_sizes = (100, )
activation = 'relu' | 'identity' | 'logistic' | 'tanh'
solver = 'adam' | 'sgd'
alpha=0.0001
max_iter = 200
early_stopping = False
validation_fraction=0.1
```

- Combination function in hidden units
  - Eg. linear combination (or weighted sum), Radial Basis Function (RBF)
- Activation function in hidden units
  - Eg. identity, sign, tanh, logistic, ReLu (Rectified Linear Unit)
- Output activation function in output unit
  - Eg. identity, softmax (for classification only)

- Multilayer perceptron (MLP)
  - the feed-forward neural network
  - linear combination function
  - ReLu/tanh/Logistic/Identity activation function

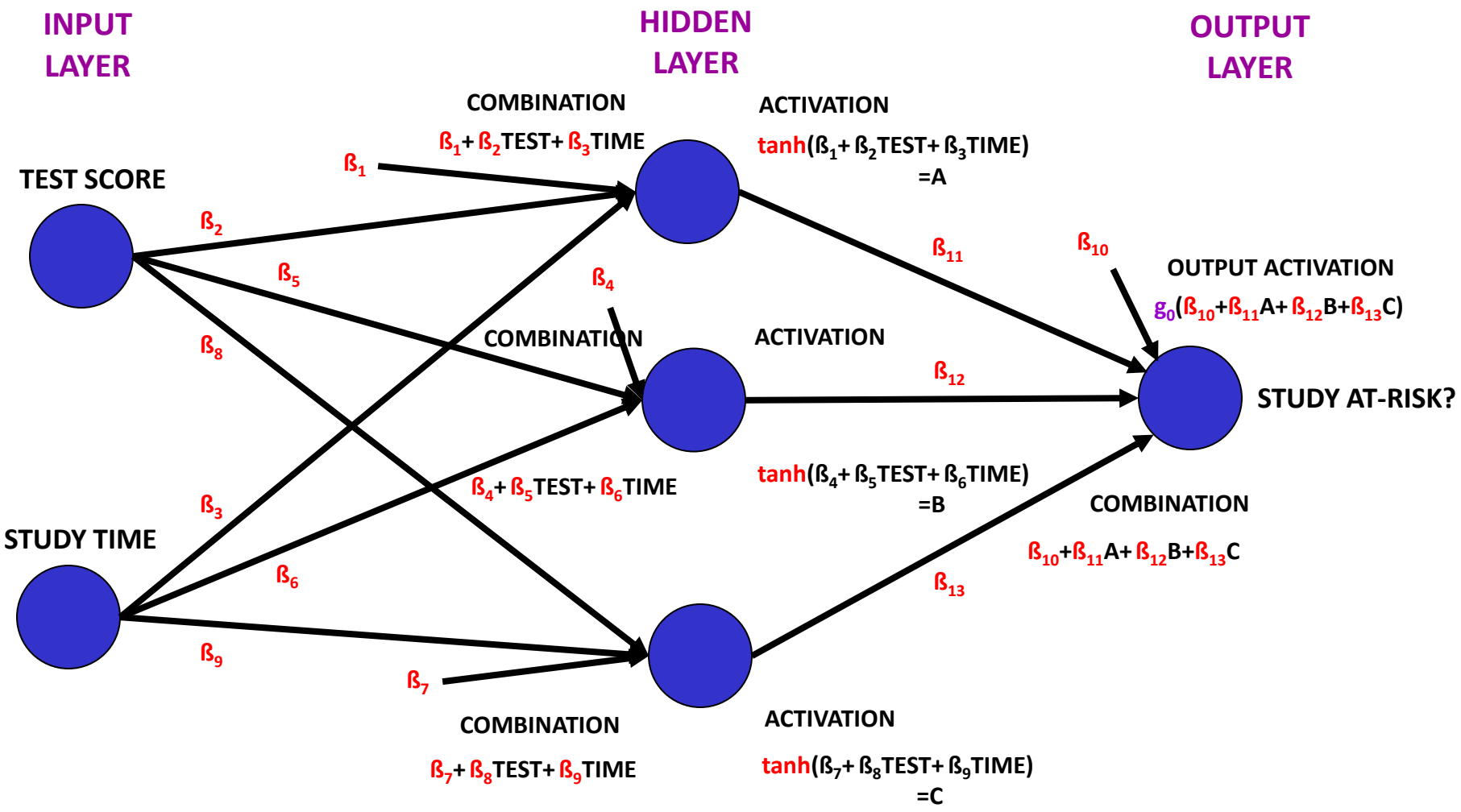




$z = w^T x + b$ Combination function

$a = \sigma(z)$ Activation function

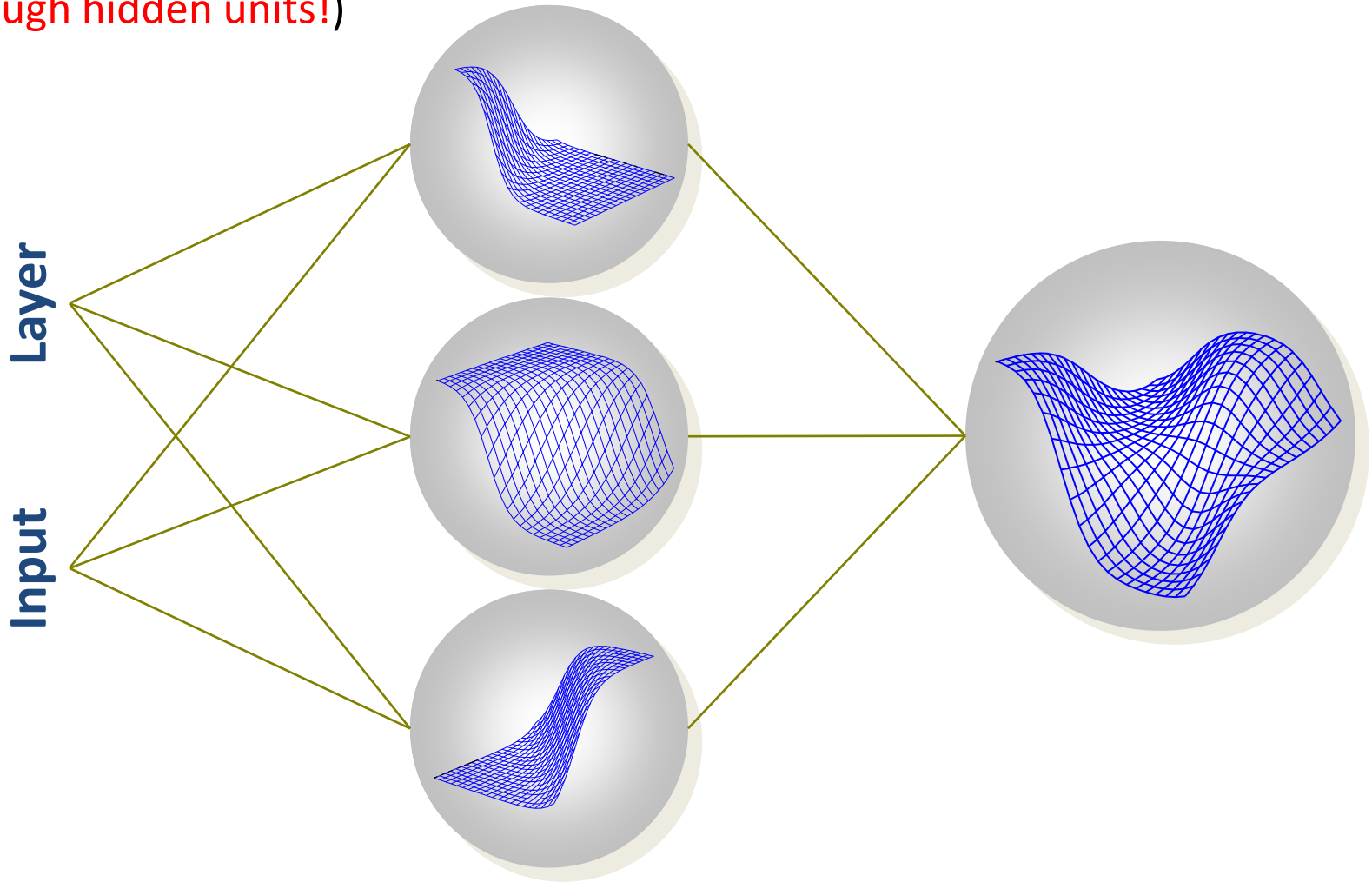
Example: MLP


INPUT LAYER
HIDDEN LAYER
OUTPUT LAYER
TEST SCORE
STUDY TIME
STUDY AT-RISK?
COMBINATION
ß1+ ß2TEST+ ß3TIME
ACTIVATION
tanh(ß1+ ß2TEST+ ß3TIME)
=A
COMBINATION
ß4+ ß5TEST+ ß6TIME
ACTIVATION
tanh(ß4+ ß5TEST+ ß6TIME)
=B
COMBINATION
ß7+ ß8TEST+ ß9TIME
ACTIVATION
tanh(ß7+ ß8TEST+ ß9TIME)
=C
OUTPUT ACTIVATION
g0(ß10+ß11A+ ß12B+ß13C)
COMBINATION
ß10+ß11A+ ß12B+ß13C
ß1
ß2
ß3
ß4
ß5
ß6
ß7
ß8
ß9
ß10
ß11
ß12
ß13

Class Activity 1:
Drawing a Neural Network

ANN: Universal Approximator

Universal approximation theorem states that a MLP with a single hidden layer can approximate any continuous function over compact domain to arbitrary accuracy, under mild assumptions on the activation function (given enough hidden units!)

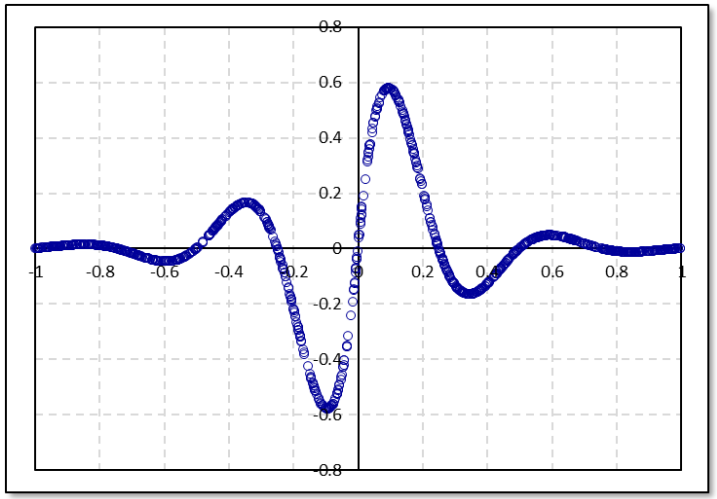


Example: Universal Approximator

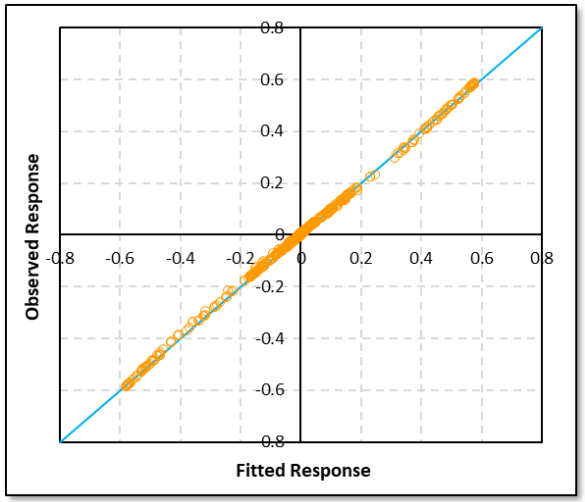
- Generate a set of 1,000 simulated data by the following function:

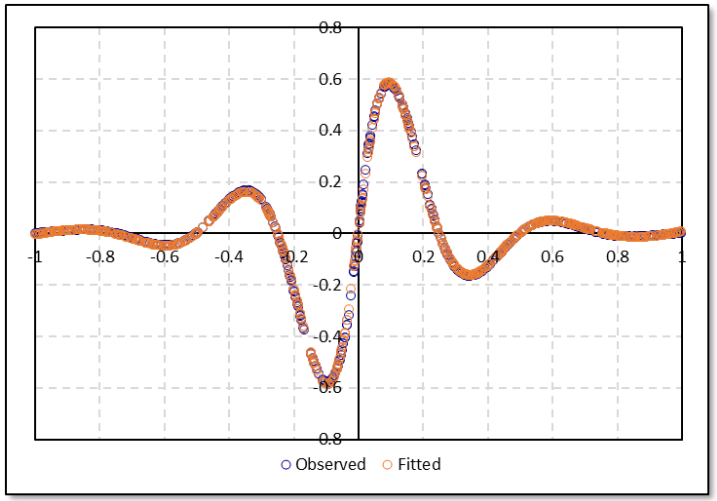
$$y = \sin(16\pi x)\exp(-5|x|),$$

where x is sampled from uniform distribution $U(-1,1)$.



- The data sets are fitted by the MLP models with 17 hidden units. Then, we have the following results:

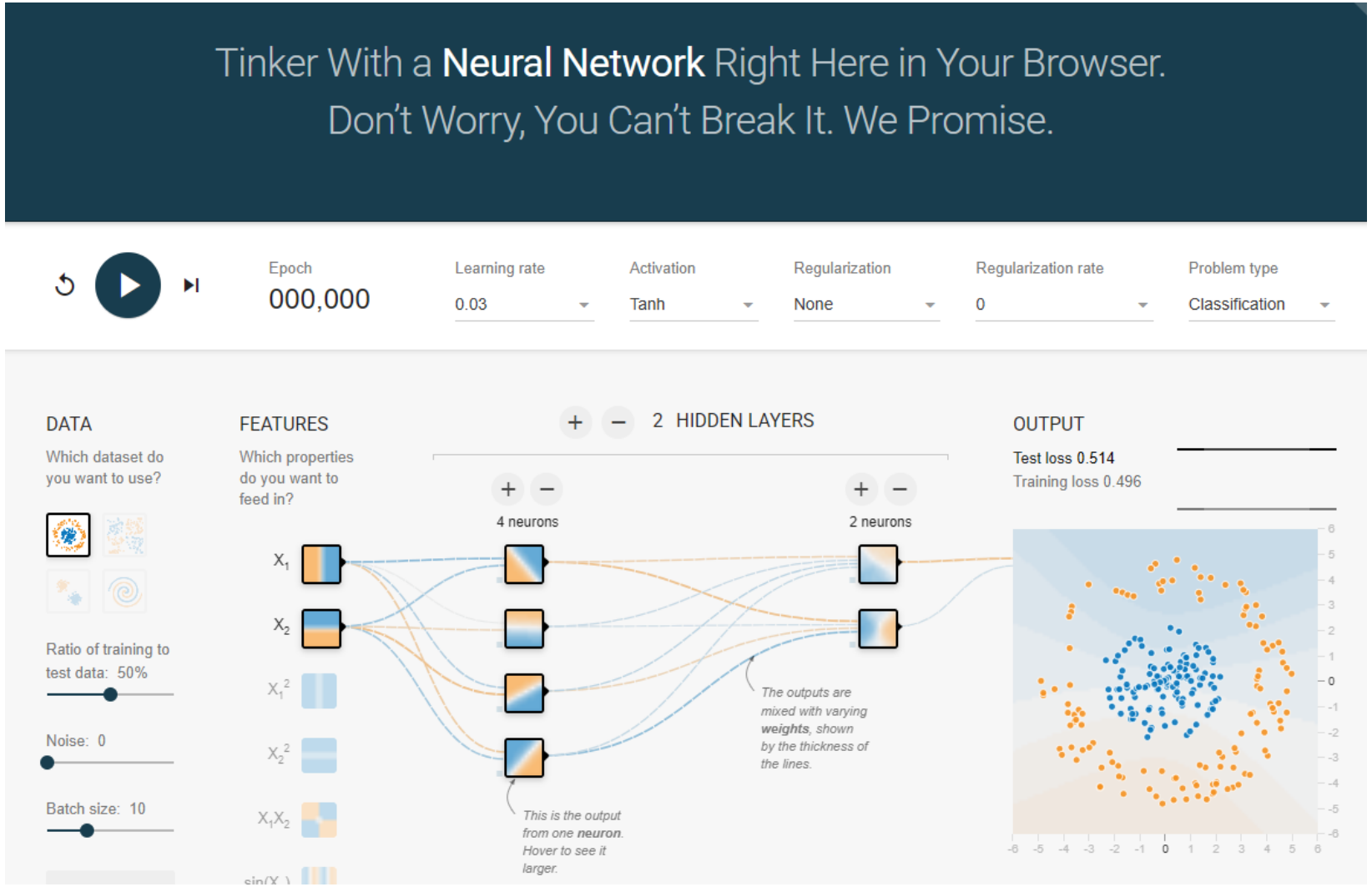




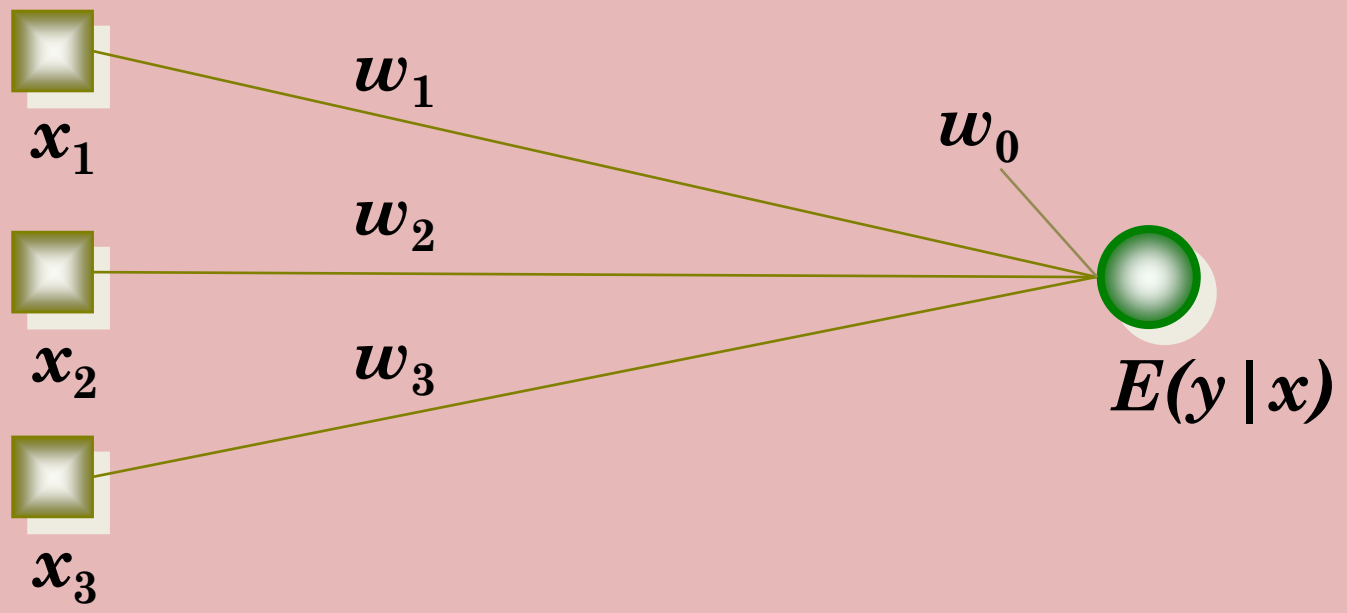
Class Activity 2:
Understanding Neural Networks with
TensorFlow Playground

TensorFlow Playground

https://playground.tensorflow.org/

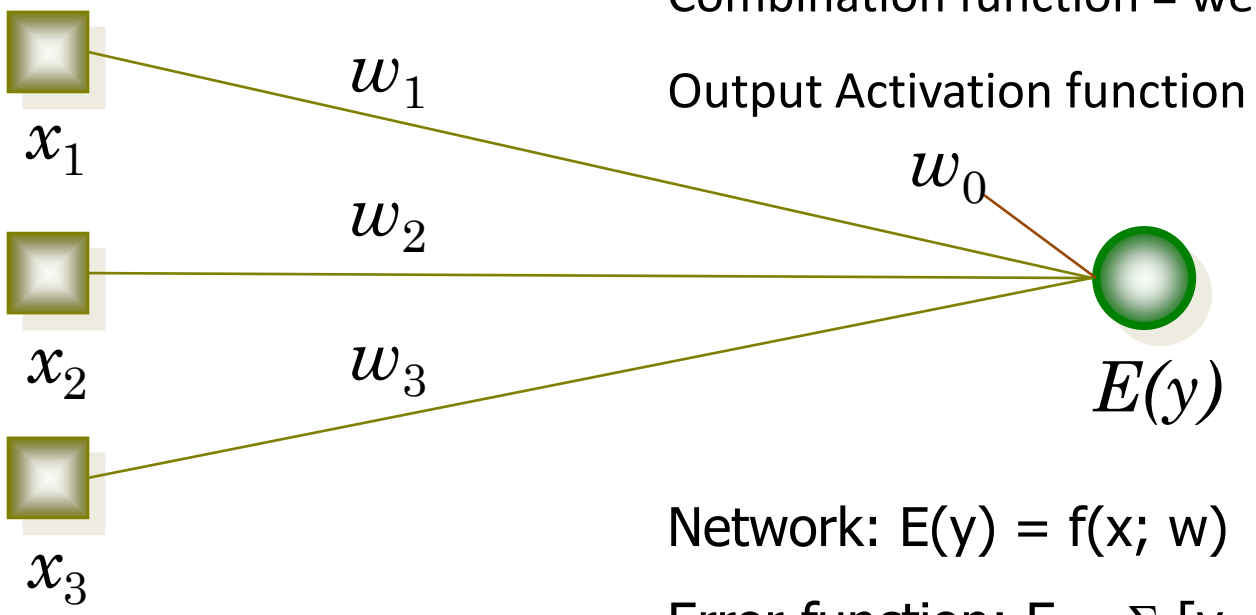


Special Cases



(Multiple) Linear Regression

$$E(y) = w_0 + w_1 x_1 + w_2 x_2 + w_3 x_3$$



Combination function = weighted sum

Output Activation function = identity

Network: E(y) = f(x; w)

Error function: E = Σ [y – f(x; w)]^2

square error

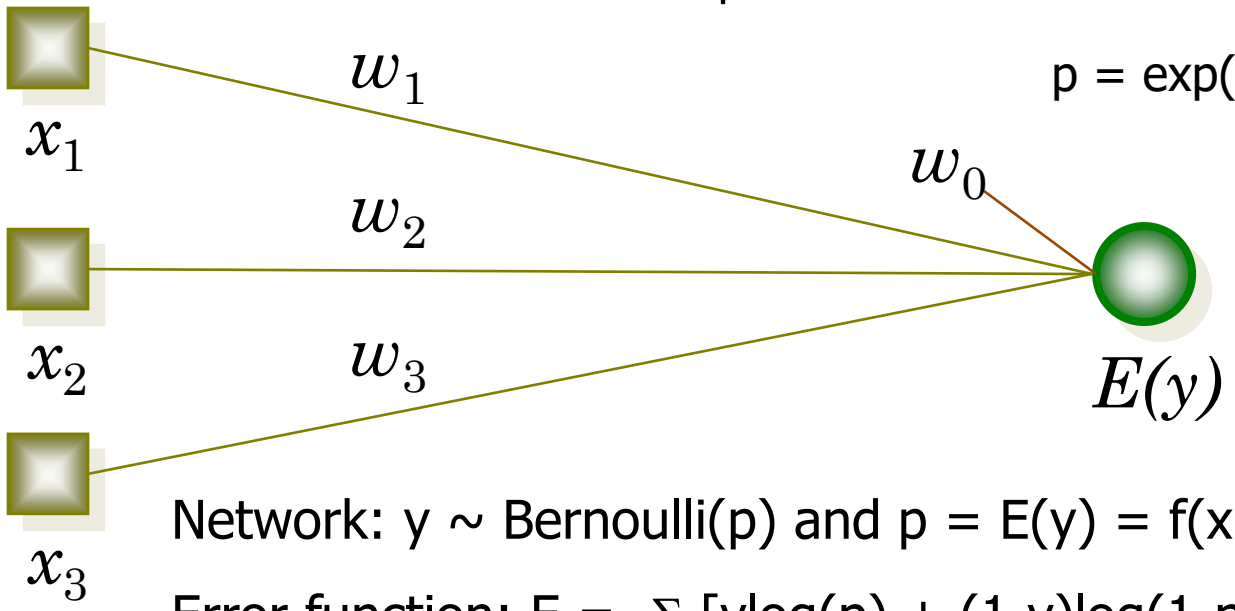
Logistic Regression

$$\ln\left(\frac{E(y)}{1-E(y)}\right) = w_0 + w_1 x_1 + w_2 x_2 + w_3 x_3 = \eta$$

Combination function = weighted sum

Output Activation function = logistic

p = exp(η)/(exp(η) + 1)



Network: y ~ Bernoulli(p) and p = E(y) = f(x; w)

Error function: E = -Σ [ylog(p) + (1-y)log(1-p)] = – log-like.

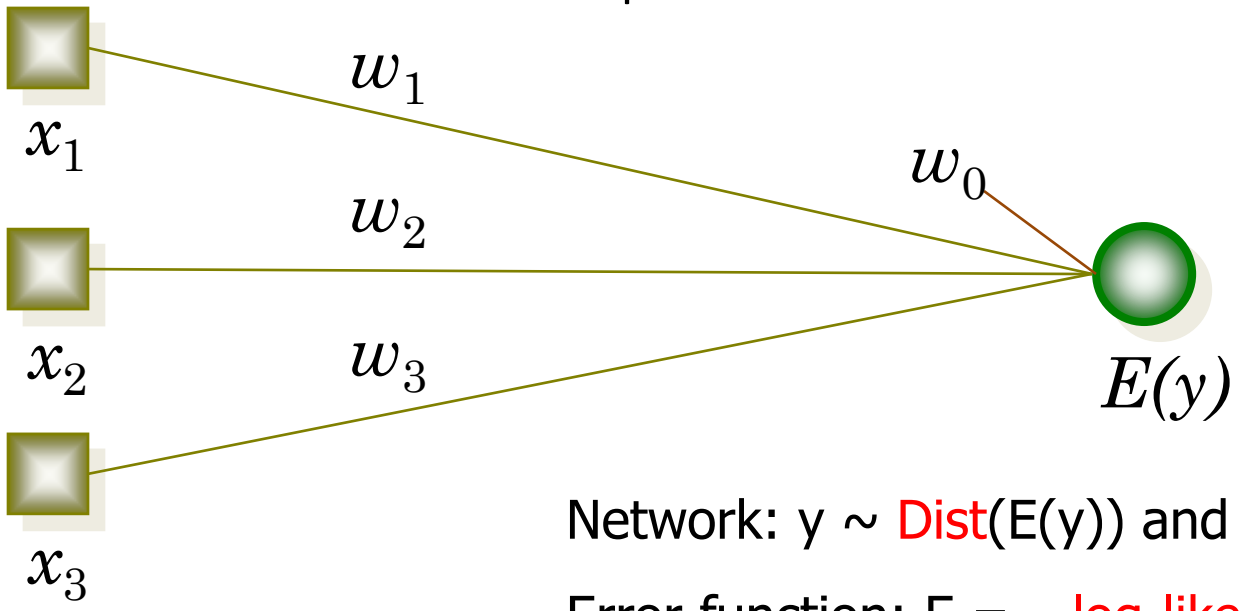
log-loss

Generalized Linear Model

$$g_0^{-1}(E(y)) = w_0 + w_1 x_1 + w_2 x_2 + w_3 x_3$$

Combination function = weighted sum

Output Activation function = inverse link function



Network: y ~ Dist(E(y)) and E(y) = f(x; w)

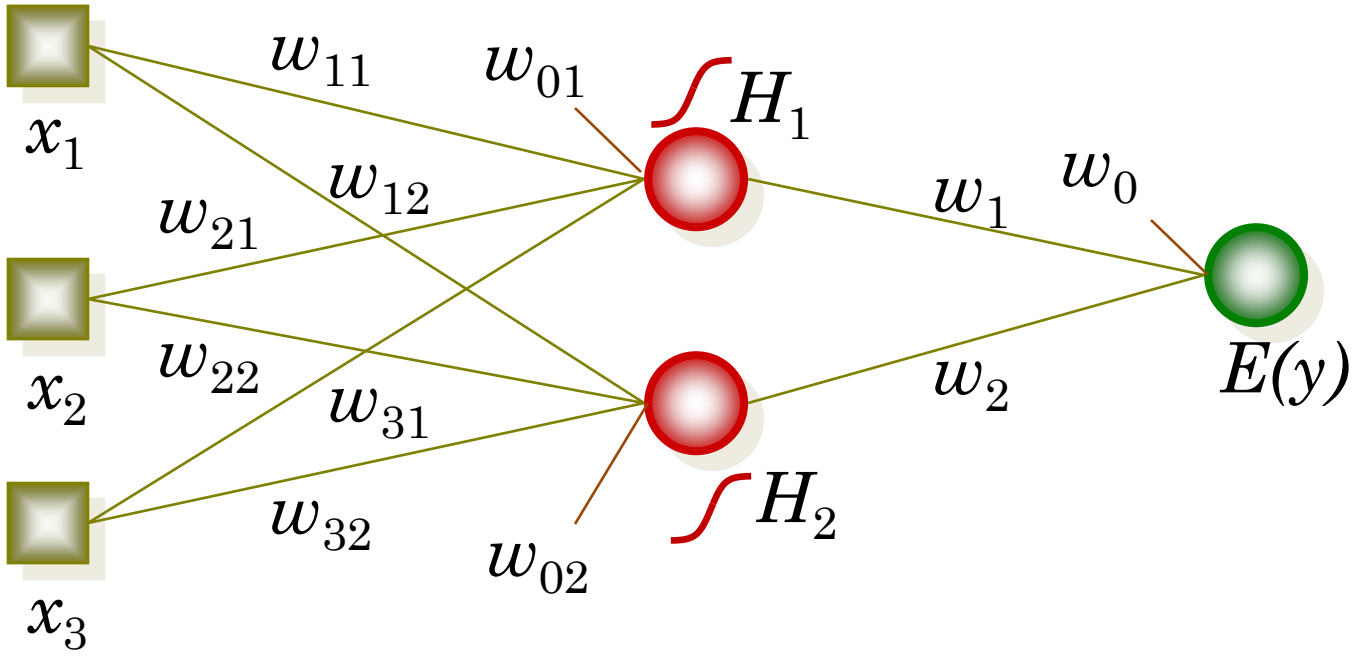
Error function: E = – log-like.

ANN with Single Hidden Layer

$$g_0^{-1}(E(y)) = w_0 + w_1 H_1 + w_2 H_2$$

$$H_1 = g_1(w_{01} + w_{11}x_1 + w_{21}x_2 + w_{31}x_3)$$

$$H_2 = g_2(w_{02} + w_{12}x_1 + w_{22}x_2 + w_{32}x_3)$$

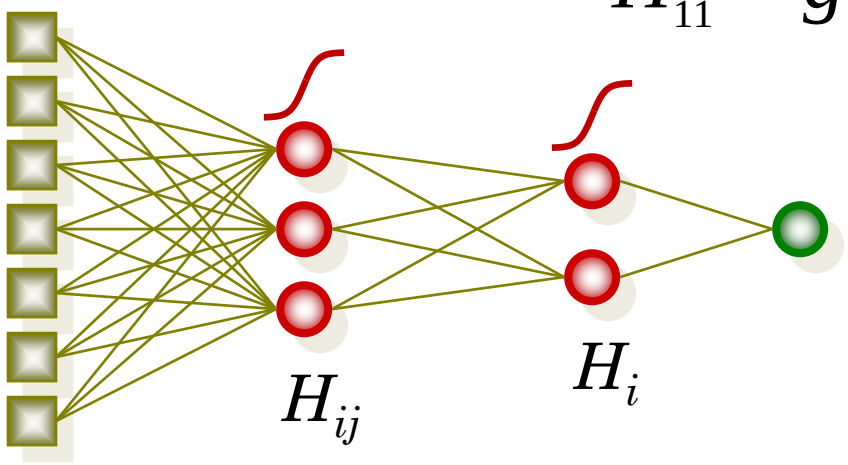


ANN with Two Hidden Layers

$$g_0^{-1}(E(y)) = w_0 + w_1 H_1 + w_2 H_2$$

$$H_1 = g_1(w_{01} + w_{11} H_{11} + w_{21} H_{21} + w_{31} H_{31})$$

$$H_{11} = g_2(w_{011} + \sum w_{i11} x_i)$$

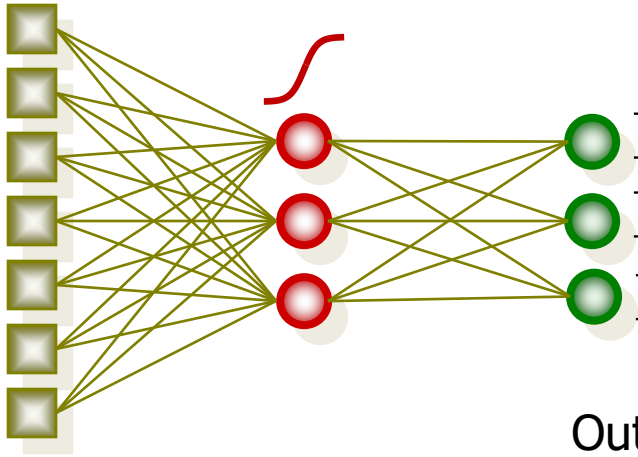


Nominal Response (Multi-Class Problem)

$$w_{01} + w_{11}H_1 + w_{21}H_2 + w_{31}H_3 = \eta_1$$

$$w_{02} + w_{12}H_1 + w_{22}H_2 + w_{32}H_3 = \eta_2$$

$$w_{03} + w_{13}H_1 + w_{23}H_2 + w_{33}H_3 = \eta_3$$



$\Pr(y = 1) = \mathrm{p}_1$

$\Pr(y = 2) = \mathrm{p}_2$

$\Pr(y = 3) = \mathrm{p}_3$

Nominal response with 3 levels

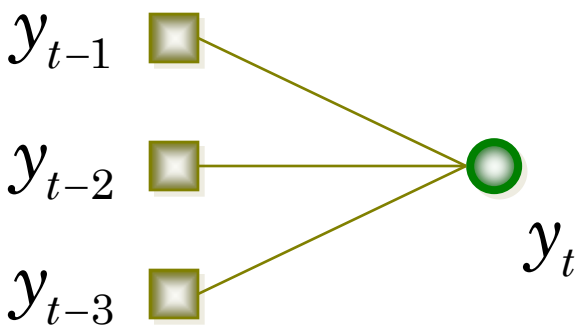
Output activation function = softmax

$$p_i = \frac{e^{\eta_i}}{\sum_{j=1}^{3} e^{\eta_j}} \qquad i = 1, 2, 3$$

(Linear) Autoregressive Model

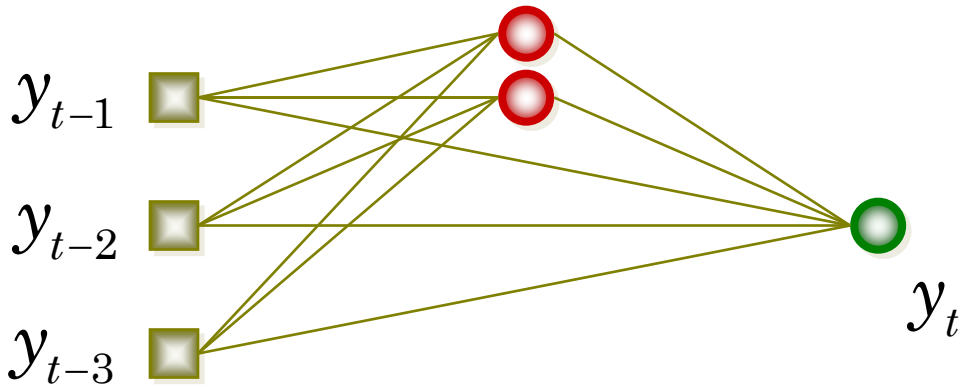
$$y_t = \phi_0 + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \ldots + \phi_p y_{t-p} + \varepsilon_t$$

AR(3)



t	y_t	y_{t-1}	y_{t-2}	y_{t-3}
1	12	.	.	.
2	13	12	.	.
3	18	13	12	.
4	19	18	13	12
5	15	19	18	13
6	8	15	19	18
7	11	8	15	19

AR-MLP



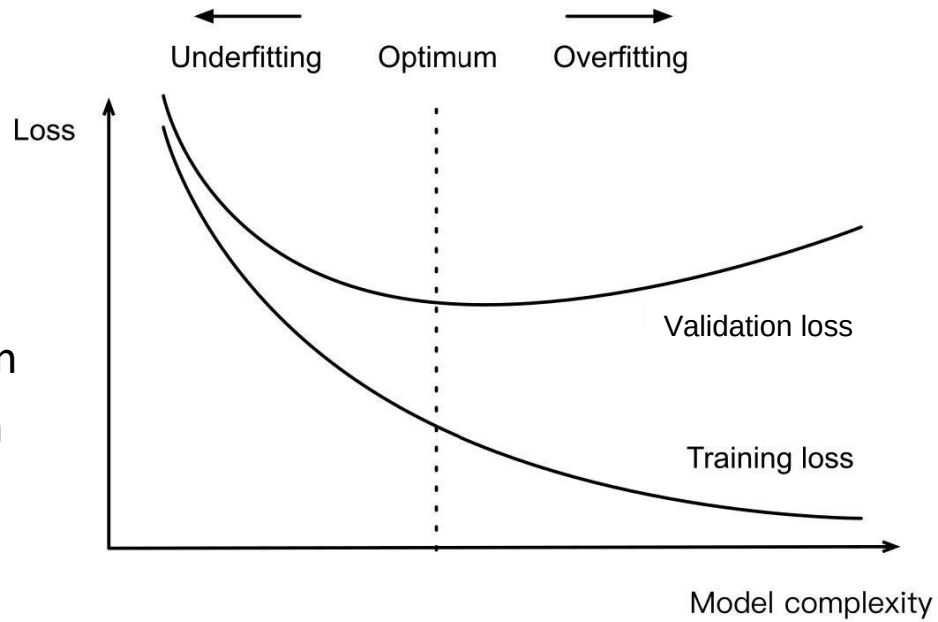
sklearn does not support direct connection

$$g_0^{-1}\left(E(y_t)\right) = w_0 \boxed{+ w_1 y_{t-1} + w_2 y_{t-2} + w_3 y_{t-3}}$$
$$+ w_4 \tanh\left(w_{01} + w_{11} y_{t-1} + w_{21} y_{t-2} + w_{31} y_{t-3}\right)$$
$$+ w_5 \tanh\left(w_{02} + w_{12} y_{t-1} + w_{22} y_{t-2} + w_{32} y_{t-3}\right)$$

Avoiding Overfitting

- As with many machine learning models, neural networks are highly prone to overfitting, where a model is able to obtain near perfect performance on the training dataset, but loses the ability to generalize to unseen data.

- Possible solutions:
  - Early stopping
  - Regularization
  - Dropout
  - Batch normalization
  - Data augmentation



According to Geoff Hinton (NIPS 2015):
"Early stopping (is) beautiful free lunch".
You should thus always monitor error on a validation set during training and stop (with some patience) if your validation error does not improve enough.

If `early_stopping = False`, then the training stops when the training loss does not improve by more than `tol` for `n_iter_no_change` consecutive passes over the training set.

If `early_stopping = True`, then it will automatically set aside 10% of training data as validation and terminate training when validation score is not improving by at least `tol` for `n_iter_no_change` consecutive epochs.

Regularization

- As with many machine learning models, neural networks are highly prone to overfitting, where a model is able to obtain near perfect performance on the training dataset, but loses the ability to generalize to unseen data.
- A common technique used to address overfitting is to do a regularization so that the overall cost function is
  - L2 regularization penalty: $J_R = J + \alpha \|\boldsymbol{w}\|_2^2 \quad \|\boldsymbol{w}\|_2^2 = \boldsymbol{w}'\boldsymbol{w}$
  - L1 regularization penalty: $J_R = J + \alpha \sum_i |w_i|$
- Choosing the right value of α is critical, and must be chosen via hyperparameter-tuning.
  - Too high a value of α causes most of the weights to be set too close to 0, and the model does not learn anything meaningful from the training data, often obtaining poor accuracy on training, validation, and testing sets.
  - Too low a value, and we fall into the domain of overfitting once again.
- It must be noted that the bias (intercepts) terms are not regularized!

Pros and Cons

- Pros
  - Handle continuous and discrete inputs and targets
  - Handle multivariate targets
  - Provide flexible and nonlinear relationship between the target and input variables
  - Smooth surface for regression
  - In general, highly accurate for fitting due to flexibility of model
- Cons
  - Model is not descriptive (black box), not easy to interpret
  - Easy to overfit the training data
  - Require imputation for missing data
  - Require to create dummy variables for categorical inputs
  - May converge to a local solution

Class Activity 3:
Interpreting a Real-World Case Study

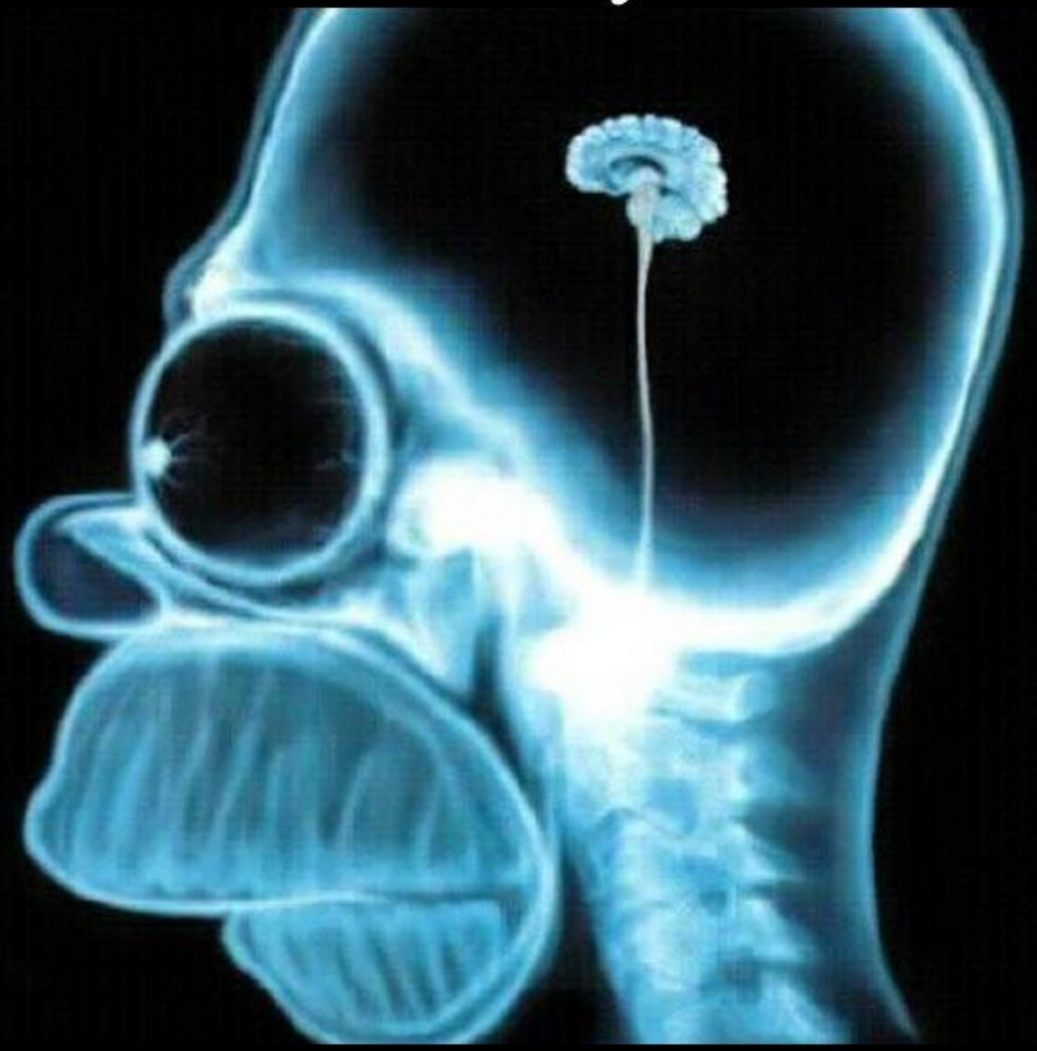
Interpreting a Real-World Case Study

- On slide 3, we have introduced 3 examples of ANN: Stock Forecasting, Soccer Prediction, Property Valuation. Think individually for 2 minutes about the following for one of the examples:
  - **What** is the input data?
  - **What** is the output being predicted?
  - **Why** is a neural network well-suited for this problem?
  - **What** is a major challenge or limitation?

Class Activity 4:

Lab 8: Accident Severity Classification

Thank you!



We hope this presentation excited your neurons.